

Abstract

The aim of this sub-project is to develop an image processing component to support low-cost automated greenhouses. The image processing application must enable the greenhouse system to establish various aspects about the plants from the images taken. These aspects include plant health, plant growth and plant disease identification from the images of different parts of the plants in the greenhouse.

What is the purpose of this section?

- Context and importance of your research.
- Outline of literatur review

Why choose image processing techniques?

- Former methods cost too much time and were expensive + destructive [1]
- Focus on image analysis component [2]
- Review plant phenotyping concepts in relation to image processing and plant breeding, discussing various measurable traits

1 Plant Phenotyping

Plant breeding is defined as the process of inducing desirable inheritable traits in plants through controlled human action [3], which is important in solving the problem of providing an adequate global food supply. The rapid growth of the human population has created a strain on international food security. It is estimated that crop production must increase by 70% by 2050 [4] to meet this demand. Additionally, abiotic and biotic stressors limit cultivar biomass and yield [5]. Therefore, plant breeders need effective techniques to produce germplasm for cultivars that maximize yield and biomass, in various adverse environmental conditions. In the last few decades, image processing techniques in plant phenotyping have been an effective tool for plant breeding [6].

1.1 What Is Plant Phenotyping?

Plant phenotyping is the evaluation of phenotypes in plants [7], where phenotypes are primitive or complex traits of an organism that can be observed or quantified and evaluated numerically [8]. Phenotypes can alternatively be seen as a combination of all morphological, physiological, chemical, and behavioural aspects that represent the individual plant [9, 10, 11].

1.2 Plant Phenotyping As Method For Improving Plant Traits

Quantitative data produced by plant phenotyping is most useful in analyzing plant-environment interactions in plant breeding [7] for the development of germplasm with improved plant traits [11]. The data produced by phenotyping is used as a proxy for assessing a plant's performance in relation to its environment. For example, stressors or disturbances of plants are induced by environmental conditions that organisms are placed in, and these stressors induce symptoms which can be detected and evaluated using phenotyping. This indicates phenotypical traits of a plant have a complex dynamic feedback loop with their environment [12]. Therefore phenotypical traits serve as proxies in determining whether certain genotypes have favourable traits in particular environmental conditions.

1.3 Image-Obtainable Phenotypical Traits As Indicators Of Plant Health, Growth, And Stress

A phenotypical trait is classifiable as morphological, physiological, or temporal. Additionally, the literature indicates morphological and physiological phenotypical traits are classifiable as holistic-based or component-based [13]. Holistic properties consider the plant as a whole, whereas component-based features look at organs like the organism's fruit, flower, leaves, stem, etc.

Leaf-derived traits are one of the most commonly used plant status indicators because leaves rapidly respond to environmental stimuli and robustly indicate various factors that determine plant health, growth and stress. [14]. Traits of leaves can monitor and predict a host of plant health, growth and stress factors. For example, leaf area predicts plant productivity because reduced leaf area results in decreased chlorophyll content and less photosynthesis. Therefore, leaf area is a good proxy measure for evaluating plant growth and has been used to monitor and estimate crop yield and health [15]. Various other leaf morphological traits exist in plant phenotyping. These include leaf number and inclination angle for monitoring plant growth and development [16, 14], and leaf thickness for measuring abiotic stressors like water scarcity [17]. Additionally, physiological traits of leaves exist that indicate various factors of plant status like stomatal conductance for the detection of biotic stressors like pathogens [18], leaf water content for drought stress [19].

Although leaves play an important role in evaluating plant growth, health and resilience to stress, these evaluations cannot be scaled to the whole plant [14]. Rather, other parts of the plant should be phenotyped to understand the processes governing plant status comprehensively. The literature is rife with such examples that study traits of roots [20], stems [21], seeds [22], and fruit [23].

2 Image Processing And Plant Phenotyping

Traditional plant phenotyping methods are destructive, prone to human error, time-consuming, and challenging to perform from start to end of a plant's ontogenesis [24, 14]. Recent advances in image sensing, processing, and data management have produced high-throughput image-based plant phenotyping systems, which are amenable to automation and can analyse large samples of data precisely in a very short period. Reducing the need for potentially erroneous, time-consuming, expensive human labour in plant phenotyping [13].

These Modern high-throughput image-based plant phenotyping systems are examples of cyberinfrastructure (CI) systems [23], which are research environments that support data acquisition, storage, management and computing and processing services distributed over some network [25]. For image-based plant phenotyping, this means that image acquisition and processing techniques are vital.

2.1 Image Acquisition Techniques In Plant Phenotyping Systems

Image processing techniques in plant phenotyping cannot be applied until suitable images are acquired. Image sensor advancements have provided the foundations for detecting various information from plants using electromagnetic radiation. Plant tissues interact with this radiation by absorbing, reflecting, emitting, transmitting, and fluorescing light. Different ranges of wavelengths correspond to different biochemical or physiological aspects of plant tissue, which are not visible to the naked eye [11]. Thus, the choice of imaging technique for data acquisition affects the types of phenotypic parameters that can be measured.

Current imaging techniques are visible, fluorescence, thermal, tomographic, and hyperspectral imaging [26, 11], where the most commonly used techniques are visible, and hyperspectral imaging due to their affordability, robustness and flexibility in measuring a wide range of phenotypic parameters.

The most common imaging technique is visible-imaging. Typically affordable, and high-resolution RGB cameras are used in visible imaging. These devices are sensitive to the visible spectral range and have been used to measure various phenotypic parameters like shoot biomass, yield and leaf morphology, and salinity tolerance [27, 28].

Hyperspectral imaging is another widely used in plant phenotyping. In this technique, sensors capture multiple spectral ranges, enabling detailed measurements for various phenotypic parameters. Notable applications of hyper-

spectral imaging include estimating nutrient content and detecting plant responses to abiotic or biotic stressors [29].

The other imaging techniques are not as widely used as visible or hyperspectral imaging; however, they are also effective in measuring plant phenotypic parameters. For instance, thermal imaging in plant water stress relations [11, 18], fluorescence imaging in used plant pathogenic symptom monitoring [30, 31], and tomography in estimating plant organ water content [32].

2.2 Image Processing Techniques In Plant Phenotyping Systems

There is a lack of generalised or standardised image-processing techniques in plant phenotyping; hence, bespoke image-processing pipelines are designed for specific plant species or genotypes like *Arabidopsis* [11, 33]. Therefore, various phenotyping systems' image processing and analysis pipelines can vastly differ; however, as mentioned earlier in the review of plant phenotyping traits, various morphological traits are derived from specific organs or parts of the plant. Furthermore, segmentation methods directly affect the accuracy of more complex tasks like object detection, 3D reconstruction, and classification. Hence, the validity and preciseness of the derived phenotypical measurements and analysis depend on the image segmentation method selected [34, 35].

In plant phenotyping, appropriate selection of segmentation techniques is vital when considering the type of phenotypic parameter to be measured. Segmentation methods in plant phenotyping are typically classifiable as traditional or learning-based [34]. Traditional segmentation methods in plant phenotyping typically target holistic phenotypical traits like plant height and include frame differencing [36], colour-based [37], edge detection [38], spectral band difference-based [39], shape modelling-based [40], and graph-based segmentation [41]. In contrast to traditional techniques, learning-based methods are more suitable for targeting component-based traits of specific organs or parts of a plant [34]. Many learning-based methods use deep learning and machine-learning mechanisms to handle segmentation [42, 43].

3 Low-cost Automated Image-Based Plant Phenotyping In Greenhouses

There is a growing need for low-cost, flexible, modifiable, and extendable automated image processing systems for plant phenotyping applications. Many automated phenotypical image processing systems in greenhouses employ various advanced technologies, such as robotics [44], automated conveyor belts, and benchtop configurations [45, 46], and advanced image sensors in high-throughput phenotyping systems [47]. However, these solutions are not accessible to all plant growers because they tend to be costly [48], and many systems are patented by specialized companies, rendering the hardware and software inflexible and unmodifiable [49]. Therefore, there is a need for low-cost, flexible, modifiable, and extendable automated image processing systems for plant phenotyping applications.

3.1 Open-source Software, Systems and Affordable Hardware As Cost Reducers In Image-Based Plant Phenotyping

Recent studies in low-cost automated image-based plant phenotyping show increased development of open-source phenotyping systems and the use of readily accessible commercial off-the-shelf hardware [50, 55] to reduce the costs of image-based phenotyping in greenhouse environments. Cheap low-cost compute sensors called Raspberry PIs have been used with affordable RGB and hyperspectral sensors to monitor phenotypic traits in greenhouse environments [51]. Additionally, there is an increase in open-source image-processing plant phenotyping systems like Phenonbox [49].

3.2 Challenges In Low-Cost Automated Image-Based Plant Phenotyping

Effective choices in image sensors and image processing techniques are crucial for developing low-cost, automated image-based phenotyping systems [14]. Modern image sensors in image-based phenotyping generate massive amounts of data [gomesComputationalSustainabilityComputational2009], necessitating the development of efficient image processing pipelines in these contexts to handle large datasets in the shortest possible time while also delivering precise and accurate phenotypical analysis [55].

The choice of sensor has a non-trivial influence on the image analysis and processing techniques used in plant phenotyping and additionally also determines the breadth of traits that can be analysed [11]. For example, hyperspectral imaging is practical for studying plant metabolic status; however, the sensors required generate Terabytes of data when monitoring multiple plants. Consequently, in low-cost settings, hyperspectral imaging of component-based

phenotypical traits becomes infeasible because learning-based image techniques, which are most suitable for this purpose, are costly, time-consuming, and impractical when large amounts of phenotypical data are processed [52].

Because modern image sensors generate a large amount of data, cost-effective data management techniques are required. Image sensors can generate large amounts of data; hyperspectral imaging can require terabytes of data to store [52], and low-cost hardware like Raspberry PIs used in automated greenhouse image processing pipelines have limited bandwidth, storage, and computing power available. Phenotyping cloud architecture systems exist [53] where compute, and data storage retrieval and uploading is delegated to, however this delegation incurs additional costs, and can potentially be cost prohibitive.

Choosing low-complexity traditional image segmentation algorithms over machine learning or deep-learning techniques can further reduce costs in image processing on low-cost hardware [52], but this also limits the number of component-based phenotypic traits that can be targeted [34], and affects the overall accuracy of measured phenotypic traits. Therefore it is important to weight the costs of savings versus accuracy, precision and breadth of measurements that can be produced [35].

3.3 Potential Advances For Low-Cost Automated Image-Based Plant Phenotyping

Existing low-cost phenotyping systems in greenhouses have bespoke image analysis and processing pipelines, which prevent them from being generalized and used in broader contexts [24, 11].

Basis for future in low-cost imaging analysis

The basis for the points below is from [25]

- Deep learning and machine learning can be used to automate image analysis, and reduce costs
- Availability of large labelled data-is bottleneck
- To use on cheap hardware you also need to optimized
- Therefore low complexity efficient algorithms for automated image pre-processing and image analysis is needed
- ref[36], one of the advantages of using ML approaches in plant phenotyping is their ability to search large datasets and discover patterns by simultaneously looking at a combination of features (compared to analyzing each feature separately). This was previously a challenge because of the high dimensionality of plant images and their large quantity, making them difficult to analyze through traditional processing methods (ref 37)

- machine learning achieved using supervised learning, so we are not able to detect novel or unexpected traits using unsupervised learning (ref 40)
- Mobile applications can be used and run processing without need for external network, cyberinfrastructure. However there are not many applications available for this [35]. Image analysis tools are often too expensive to run on these devices due to the constrained resources, however they are much cheaper and therefore can be exploited. In particular low complexity efficient systems, which generalize to various genotypes, and phenotypic measurements are needed. In particular the development of efficient low resource intensive machine modelling and deep learning techniques can be made. [35]. In developing low-cost image analysis and preprocessing tools for phenotyping it is necessary to weight the cost savings versus accuracy and precision. But bespoke image processing techniques do not lend well to generalizability and extendability.
- automated systems not built for general solutions but for specific species of plants [7]

Current applications

- Often combined with machine learning algorithms
- Cheap Raspberry PI with RGB camera [51]
- use off the shelf commercial software like Microsoft Kinect [50]

What affects a low-cost automated greenhouse cost

Points from this paper [54]

- Image preprocessing pipeline
- open source software
- hardware
- amount of labour and time needed for analysis
- energy cost of technology
- amount of data generated, where is it stored, how frequently? [2]

Challenges of low-cost automated greenhouses

- Many phenotyping systems existing are commercially developed under patent so hardware and software can not be modified. [47]
- Current applications use moving conveyor belt (benchtop) [47]
- or conveyer belt [47]

- or moving robotic arm to move sensors around to get high-throughput measurements.
- large amounts of data need to be collected, and processed. Existing phenotyping platforms exist, but they are expensive and have restricted running environments.
- Many systems developed for laboratory systems, but not extended for real applications. Standalone off the shelf equipment, open-source software allow modular and extensible plus affordable solutions [55]
- Custom data collection schemes need to be made very little standards
- lack of robust methods to analyse plant images
- low cost moving track systems limited in phenotypical, genotype and management types are supported [56] suggesting need for low-cost systems that can support a wider variety of genotypes and phenotypical traits.

Fill out the conclusion section.

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